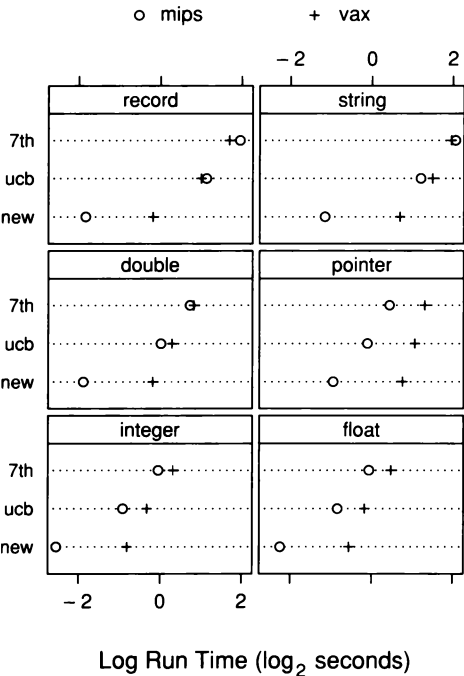


the machine factor, we could assign it to the panel levels, but Figure 6.13 uses another method: superposition. The machine factor is encoded by the symbol type. Now its effect can be assessed for each combination of algorithm and input by comparing the two symbols on each panel level. The symbol encoding succeeds because there are only two machine categories. Superposition is used again in Figure 6.14, where the input factor is assigned to the panel levels, which allows better assessment of the effect of input.

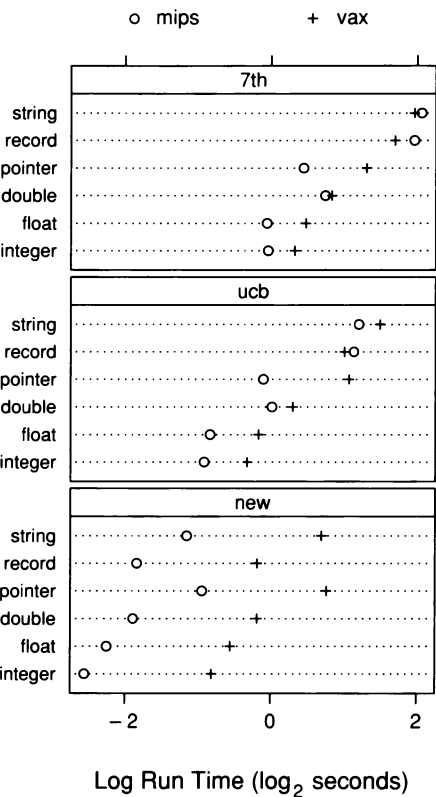
The decrease in run time of the Bentley-McIllroy algorithm is impressive. Figure 6.12 shows that it beats Seventh Edition and Berkeley in every case. The improvement gets as large as $3.8 \log_2 \text{ sec}$, a factor of 14. The figure also shows that the size of the improvement depends on the machine and the input. The size is greater for the Mips than for the Vax. Pointers stand out as a special case; on each machine, the log run times for pointers are less spread out than for any other input.



6.13 The log run times are graphed by a multiway dot plot with superposition.

Figure 6.13 shows that the effect of the machine factor for the new algorithm is much greater than for the other two. Furthermore, the machine effect for the new algorithm is stable; the log run time for the new algorithm on the Vax minus the log run time on the Mips, is always close to about $1.75 \log_2 \text{ sec}$. For the other two algorithms, the machine effect depends on the input type; it is close to zero for record, string, and double, but is larger for other inputs.

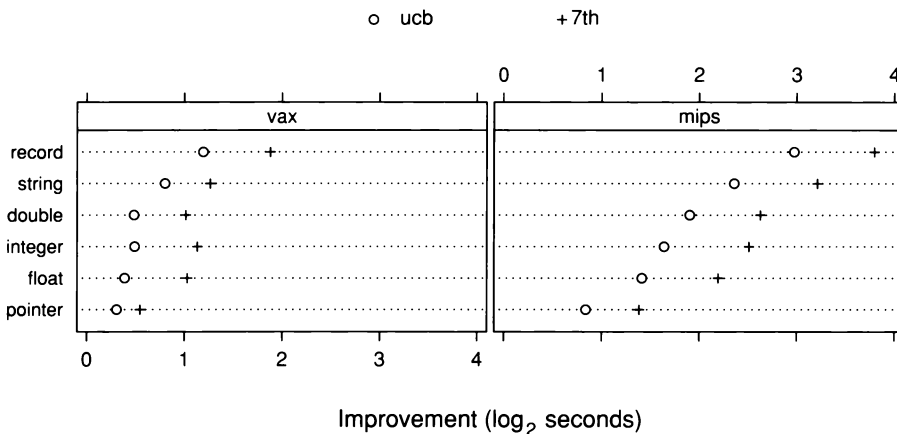
Figure 6.14 shows the patterns for Seventh Edition and Berkeley have similar shapes that appear to be separated by an additive shift of Berkeley toward smaller log times. In other words, Seventh Edition and Berkeley behave very similarly except that Berkeley is somewhat faster by a constant number of $\log_2 \text{ sec}$.



6.14 The log run times are graphed by a multiway dot plot with superposition and with the input type encoded by the panel levels.

Differences

For the log run-time data, the categorical variable of chief interest is the algorithm. The experiment was not carried out to compare the speed of sorting on Vax and Mips machines or by input type. Machine and input enter only because the relative performance of the different algorithms depends on them. Furthermore, we are primarily interested, not in the comparison of Berkeley and Seventh edition, but rather in the magnitude of the improvement provided by the new algorithm. We can focus on this improvement by visualizing a new data set derived from the old — differences of log run times. The differences are graphed in Figure 6.15. The circles display the Berkeley log times minus the new log run times for all combinations of machine and input. Similarly, the plus signs display the Seventh Edition log times minus the new log times. The levels and panels have been ordered by the category means of the differences. Means have been used instead of medians since the exploratory visualization showed the data to be well behaved.



6.15 Each value graphed on this multiway dot plot is a Vax or Mips log run time minus the log run time of the new algorithm for the same input and machine.

On each panel of Figure 6.15, the overall patterns of the two sets of points are about the same, but simply shifted. In other words, for each machine, the data are additive. But we do not have such additivity across machines; the patterns for the Vax are quite different from those for the Mips, although the order of the values for each of the four cases — Vax-7th, Vax-Berkeley, Mips-7th, Mips-Berkeley — is nearly the same.

Figure 6.15 suggests that we fit a function that is additive in input and algorithm for each machine separately. In other words, in the fitting, we treat the data for each machine separately and fit an additive function in each case. Let m index the two machines, let i index the six input types, and let a index the two algorithms. The form of the function is

$$\mu^{(m)} + \alpha_i^{(m)} + \beta_a^{(m)} .$$

For each of the two machines, the input main effects

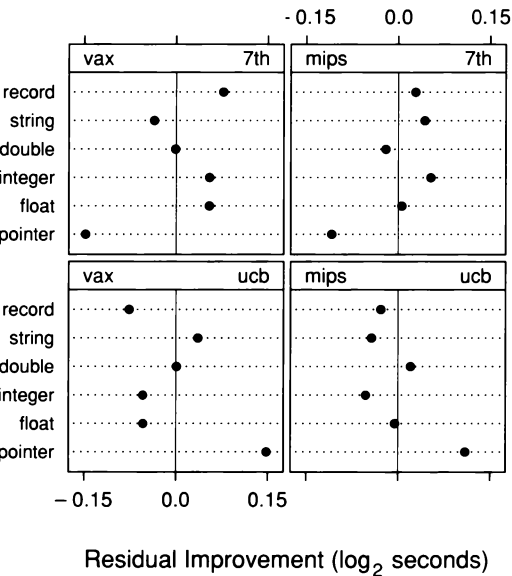
$$\alpha_i^{(m)}$$

sum to zero and the algorithm main effects,

$$\beta_j^{(m)}$$

sum to zero. Since the data appear to be well behaved, we will not employ bisquare in the fitting.

Figure 6.16 graphs the residuals; no strong effect of the variables appears to remain. In other words, there is no discernible lack of fit, so the function fits the data.

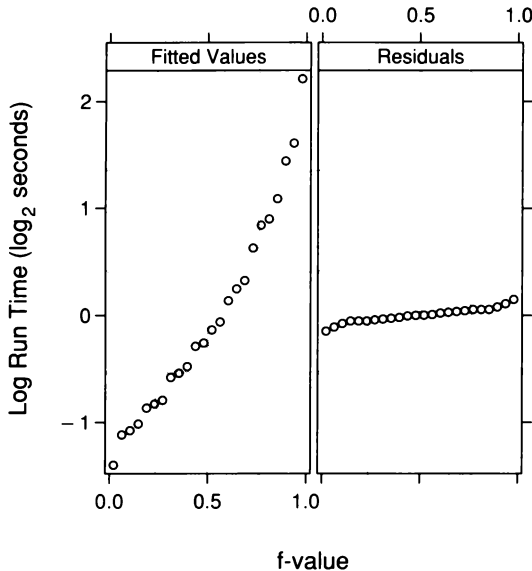


6.16 The multiway dot plot graphs the residuals from the fit to the differences.

Figure 6.17 is an r-f spread plot. The residuals are contained within $\pm 0.15 \log_2 \text{ sec}$, which means that on the original scale, the fitted function predicts run-time improvement to within factors of

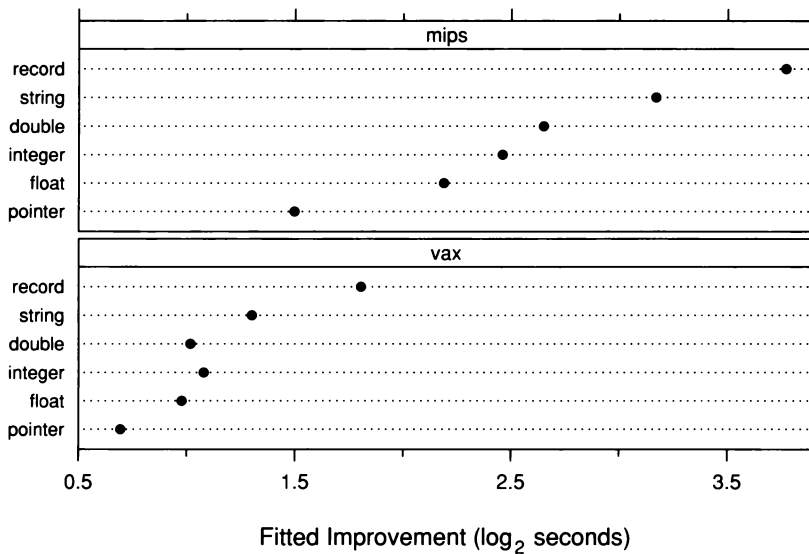
$$2^{\pm 0.15} = 1 \pm 0.1 .$$

Thus the fitted function accounts for most of the variation in the data.

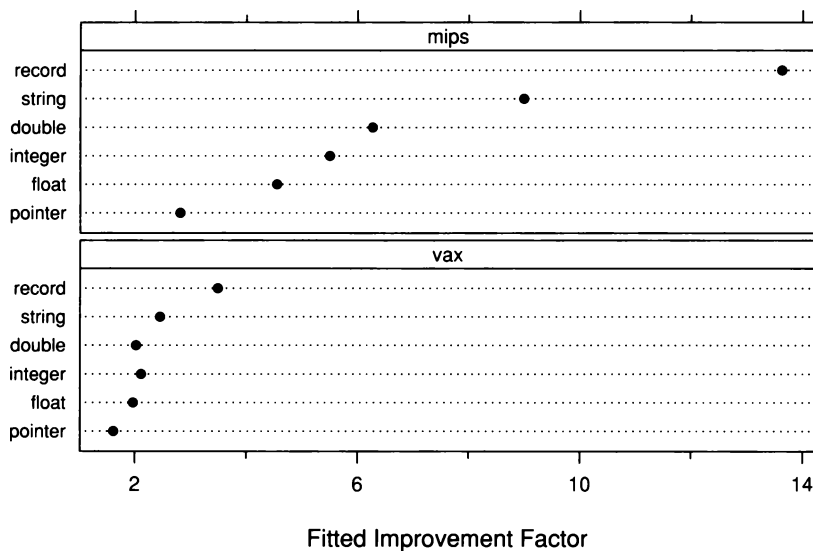


6.17 The r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the fit to the log run-time differences.

The additivity for each machine allows a simpler characterization of the improvement. For the Mips, the algorithm main effects are $-0.38 \log_2 \text{ sec}$ for Berkeley and $0.38 \log_2 \text{ sec}$ for Seventh Edition. Thus the change in the Mips fit for any input type in going from Seventh Edition to Berkeley is $-0.76 \log_2 \text{ sec}$, which is a factor of $2^{-0.76} = 0.6$. In other words, on the Mips, the improvements over Berkeley are 0.6 times the improvements over Seventh Edition. A similar statement holds for the Vax with a factor of 0.7. Thus we need only consider in detail the improvement of the new algorithm over either Berkeley or Seventh Edition to understand the effect of machine and input. This is done in Figure 6.18, which displays the fit for the improvement over Seventh Edition on the $\log_2$ scale, and in Figure 6.19, which displays the fit as a factor on the original scale.



6.18 The fitted values for Seventh Edition are displayed.



6.19 The fitted values for Seventh Edition are displayed as improvement factors.

6.4 *The Case of the Anomalous Barley Site*

The book ends as it began, with the barley data. There is a mystery — anomalous data at the Morris site. Its unraveling, to come next, provides strong testimony to the themes stressed in the book.

The Barley Data

Let us review the facts of the data. Ten varieties of barley were grown at six sites in Minnesota in each of two years, 1931 and 1932. Thus there are $10 \times 6 \times 2 = 120$ yields. The data first appeared in 1934 in the *Journal of Agronomy* [55].

The story of the data actually begins earlier, in the 1920s in England, where R. A. Fisher was establishing the foundations of statistical experimental design and modern statistical inference [38, 39]. Fisher was a brilliant scientist who also was a pioneer of mathematical genetics. His work in statistics would go on to permeate all of science and have a profound effect on the everyday lives of scientists.

Fisher's work in statistical experimental design began at the Rothamsted Agricultural Experiment Station, where he was employed for over a decade before he took the post of Galton Professor at University College London in 1933. Fisher revolutionized how people approached experiments. When he began his work, the prevailing wisdom was that only one factor should be varied at a time in a single experiment to keep investigations simple. Fisher demolished this notion. He showed by a combination of empirical and mathematical work that simultaneous variation of the levels of several factors in certain systematic ways in single experiments is far more efficient. Even more, Fisher showed how to assess the precision of estimates of factor effects. His method, now a mainstay of science, is called the *analysis of variance*, or *ANOVA*. Yates and Mather wrote in 1963 [82]:

... the new ideas on experimental design and analysis soon came to be accepted by practical research workers, and the methods have now been almost universally adopted, not only in agriculture but in all subjects which require investigation of highly variable material. The recent spectacular advances in agricultural production in many parts of the world owe much to their consistent use.

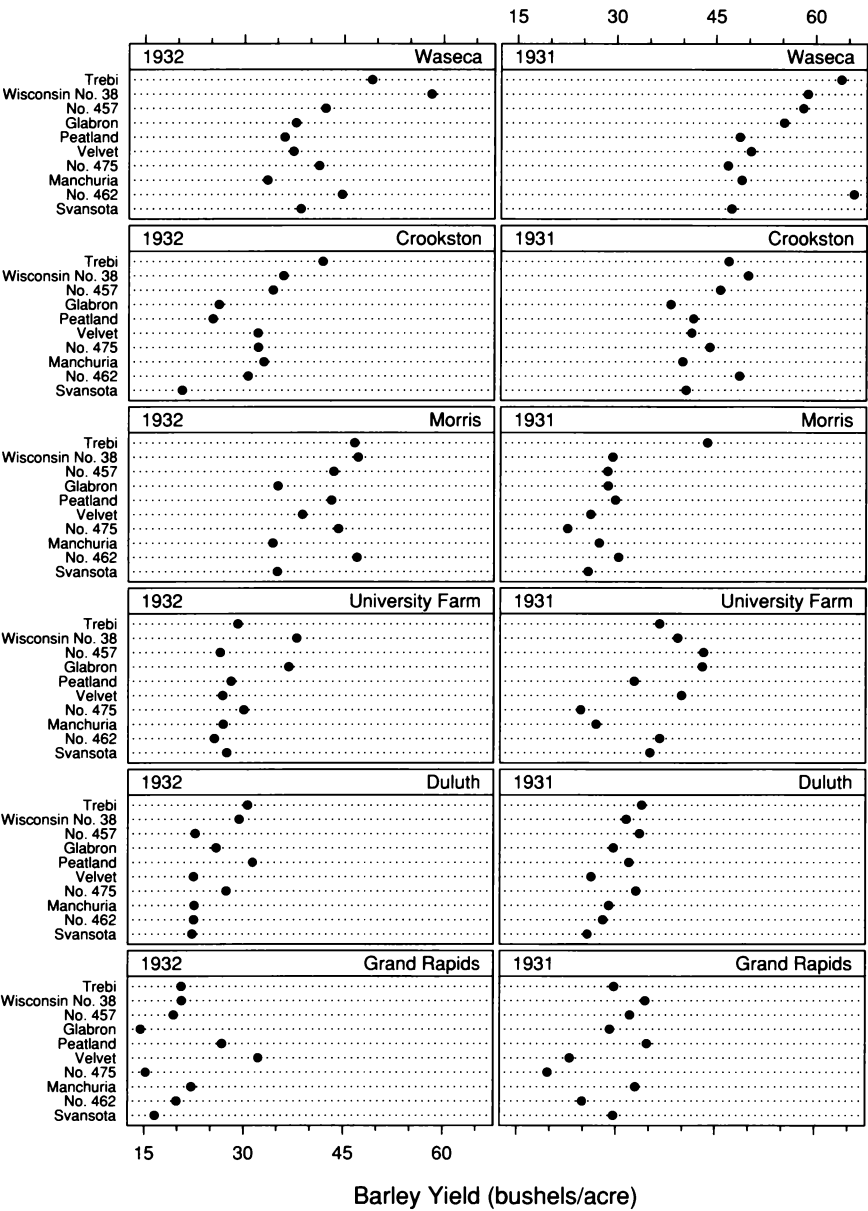
The barley experiment was significant because it was one of the first uses of Fisher's ideas. Fisher later reported the data from five sites, including Morris, in *The Design of Experiments* [39], which became the classic in experimental design, setting out the foundations of the subject. The fame of the book made the data famous, and they were re-analyzed by others, serving as an example for new methods of statistical science.

Exploratory Dot Plots

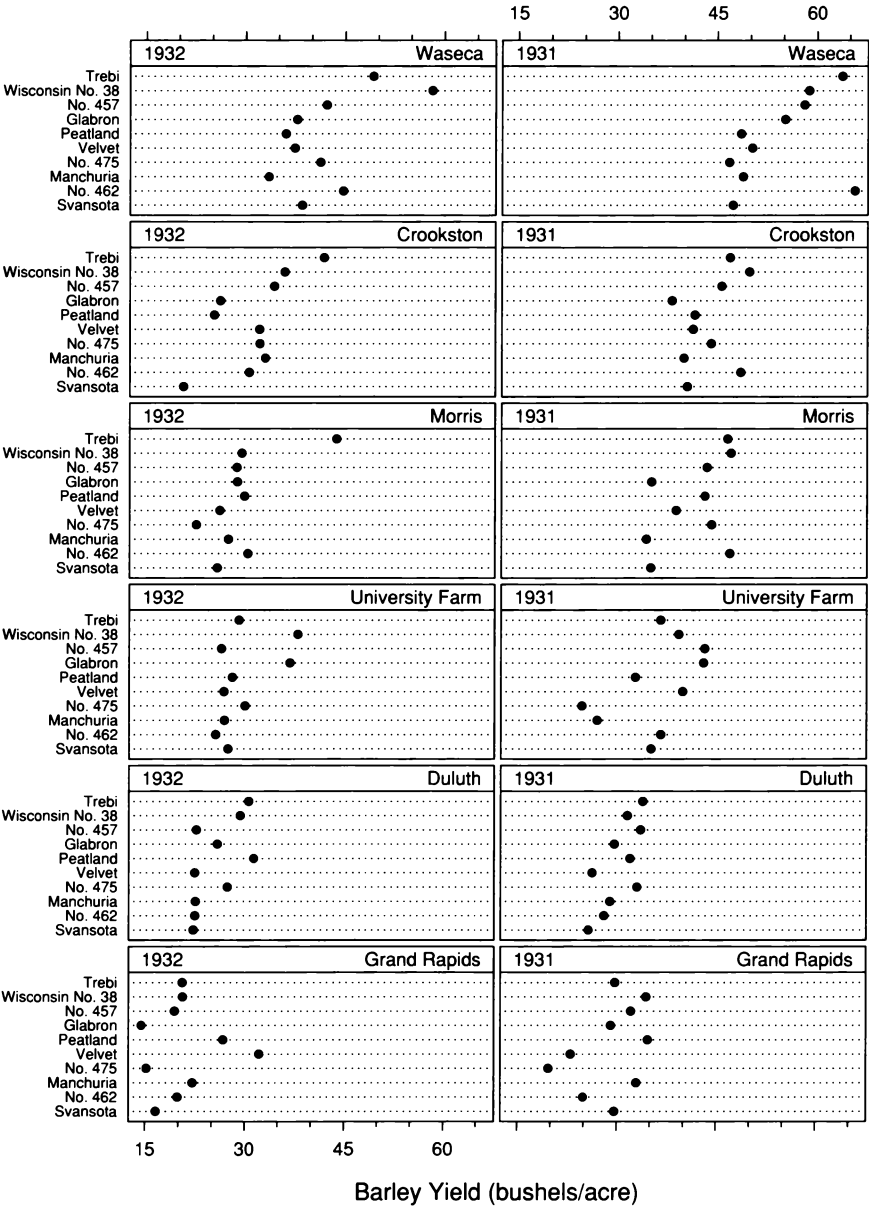
Figure 6.20 graphs the barley data by a multiway dot plot. The variety variable is assigned to the panel levels, the growing site is assigned to the panel rows, and the year is assigned to the columns. The order of the levels and panels is by category median. The variety medians increase from bottom to top through the levels, the year medians increase from left to right through the panel columns, and the site medians increase from bottom to top through the panel rows.

Figure 6.20 clearly shows the Morris anomaly: a reversal of the yearly effects. But the display also conveys a pattern that is crucial to understanding the reversal. As we proceed from bottom to top through either column of site panels, the overall levels of the sites increase, except for Morris. Furthermore, the two patterns without Morris are roughly the same except that 1931 is shifted toward higher values. Finally, and most importantly, the visual impression is that if the years for Morris were interchanged, the Morris data would then fit the pattern.

The data are shown with the interchange in Figure 6.21. Now Morris does indeed fit the pattern. In other words, given the overall median of Morris, its 1932 values fit into the 1931 pattern of the other sites, and its 1931 values fit into the 1932 pattern. The interchange has produced greater harmony in the data. But this is a serious step, not to be taken lightly. We will persist in analyzing the altered data, assessing the effect of the alteration and studying further whether it is justified.



6.20 The multiway dot plot graphs barley yield at six sites for two years and ten varieties.



6.21 The barley data are displayed again with the 1931 and 1932 yields at Morris interchanged.

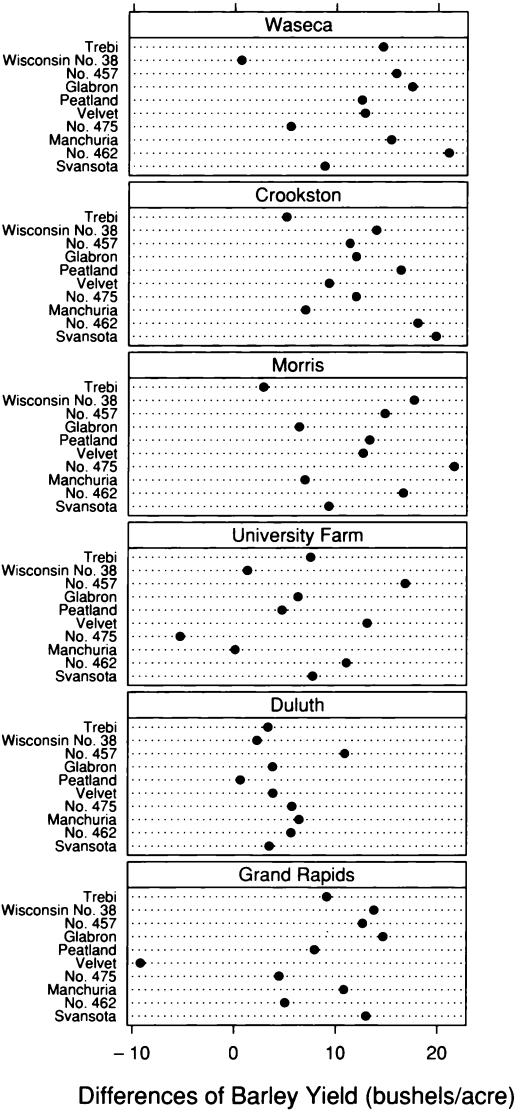
Visualizing the Altered Data

The chief issue at the moment is the year effect. The situation is similar to that of the previous section where the categorical variable targeted for study was the algorithm factor. Here, as there, we will take differences to focus on the targeted variable.

Figure 6.22 displays the 1931 yield minus the 1932 yield for each combination of site and variety. The order of the sites and varieties remains as in previous displays, so as we go from bottom to top through the panels, the median site yield increases, and as we go from bottom to top through the levels on each panel, the median variety yield increases. This allows us to assess whether there is a relationship between the differences and the site medians or between the differences and the variety medians. One noticeable effect on the display is that from bottom to top, the overall level of the differences at each site tends to increase. In other words, the effect of year at the different sites tends to increase as the overall site level increases. There is an interaction between site and year. The only site that appears to deviate from this pattern is Grand Rapids, not Morris. A similar interaction between variety and year does not appear to be present; no panel shows any tendency for the differences to increase.

An increase in the effect of one factor as the overall level of another factor increases is an exceedingly common form of interaction. Thus the interaction between site and year uncovered by Figure 6.22 is not surprising. Morris fits right into the pattern of this interaction. Thus the interchange of the years appears to be striking an even more harmonious chord than we previously expected.

But Figure 6.22 reveals new aberrant behavior. There is an unusually low value for Velvet at Grand Rapids, about -9 bushels/acre. In other words, the 1932 Velvet yield exceeds by a large margin the 1931 Velvet yield. We will resist the temptation to interchange the years, but the behavior of the two values makes it clear that for our fitting, coming next, we will need bisquare.



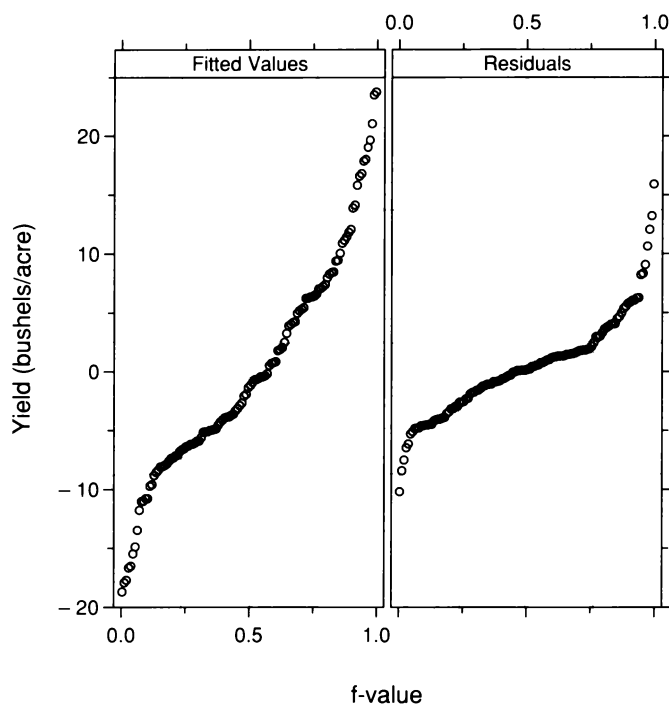
6.22 The multiway dot plot displays the 1931 yields minus the 1932 yields for the altered data.

Fitting the Altered Data

The fit to the altered yields must account for the interaction between year and site. This is done by the function

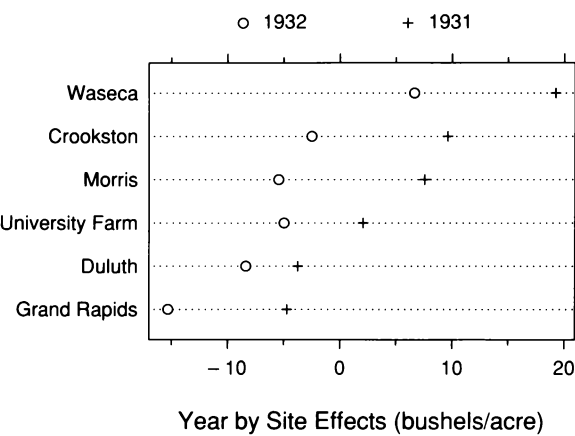
$$b_{vys} = \mu + \alpha_v + \beta_{ys},$$

where v indexes the variety, y indexes the year, and s indexes the site. The 10 variety main effects, α_v , sum to zero and the 12 year-site interaction effects, β_{ys} , also sum to zero. Again, we will use bisquare to do the fitting. Figure 6.23 is an r-f spread plot; the fitted function accounts for much of the variation in the data.

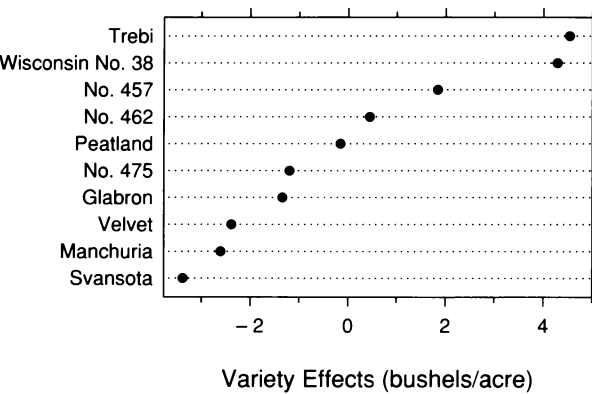


6.23 The r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the bisquare fit to the altered data.

The bisquare estimates of the year-site effects are shown in Figure 6.24. The pattern of the interaction between year and site, spotted in the data, is also evident here; as the mean of the two interaction effects for each location increases, the difference tends to increase. Figure 6.25 graphs the estimates of the variety effects. Amidst our interest in the Morris anomaly, it should not be forgotten that the original goal of the experiment was to investigate the performance of the different barley varieties. Trebi and Wisconsin No. 38 are the clear winners. Svansota is the loser. The range of the variety main effects, -3.4 bushels/acre to 4.5 bushels/acre, covers 7.9 bushels/acre. The range of the year-site interaction effects is considerably greater and covers 34.5 bushels/acre, which is a factor of about 4.5 greater. Thus, while selection of the best variety can improve yield, the changes in growing conditions that occur with changes in geography and time are clearly far more influential.

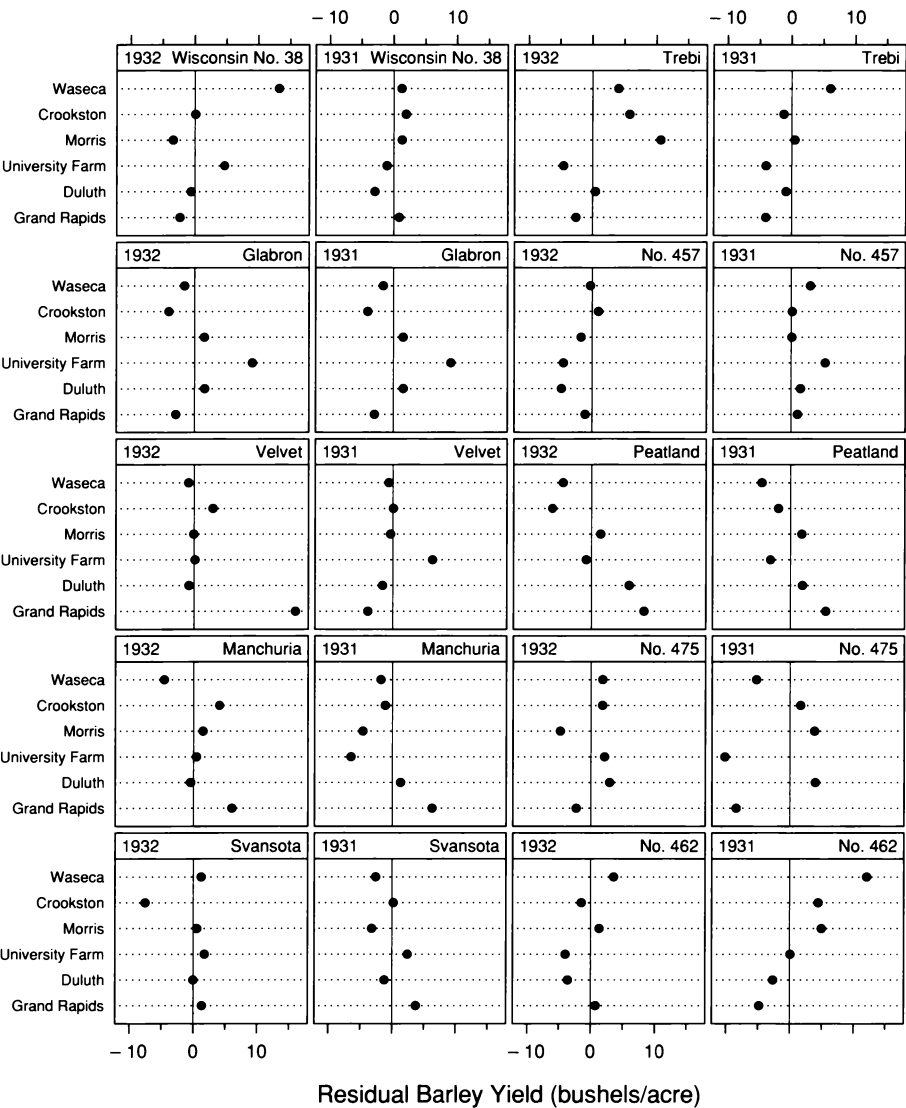


6.24 The dot plot graphs the bisquare estimates of the year-site interaction effects for the altered data.

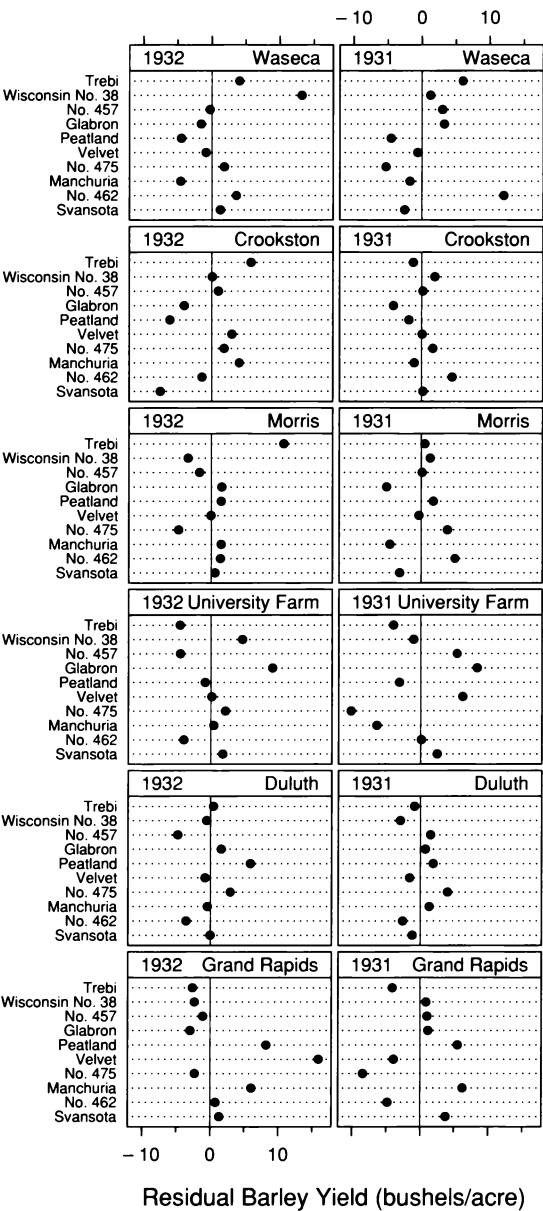


6.25 The dot plot graphs the bisquare estimates of the variety main effects for the altered data.

Figures 6.26 and 6.27 graph the residuals. We have not bothered to re-order the panels by the estimated effects, maintaining the category-median ordering of the exploratory visualization of the data. The reason is that no consistent pattern appears. The relatively simple fitted function appears to fit the data satisfactorily.



6.26 The multiway dot plot displays the residuals from the bisquare fit to the altered barley data.



6.27 The multiway dot plot displays the residuals from the bisquare fit to the altered barley data.

Verdict

A simple function fits the altered data. No discordance has resulted from the Morris interchange. The harmony and simplicity that result from the alteration force us to entertain the possibility that the data are in error. The alternative is that an extraordinary natural phenomenon occurred — the yields at Morris happened to be reversed by nature in such a way that the negative of the Morris year effect just fits into the pattern of the year-site interaction at other sites. On the basis of the evidence, the mistake hypothesis would appear to be the more likely.

But there is other evidence, from the agronomists' report, that adds to the likelihood of the mistake hypothesis. This clue was discovered by Francis J. Anscombe [3]:

There is one puzzle in the paper, namely that the crops are first described as having been grown in 1930 and 1931, but thereafter in all tables and discussion the years are referred to as 1931 and 1932. The same discrepancy occurs in Fisher's book. Presumably the later dates are correct. Barley in Minnesota is apparently always sown in the spring, so that the whole growing process occurs in one calendar year.

The incorrect specification of the years in the report evidences a carelessness that could easily infect other aspects of the specification of years.

What we need now is something akin to a motive. How could three agronomists analyzing data from the very state in which they lived overlook a blunder of this magnitude? The answer is provided by two sentences in their report:

This paper is essentially a methodology study and illustrates the application of the analysis of variance to an endeavor to obtain answers to the above questions [statistically significant differences in the effects of the three categorical variables]. The statistical method used, known as the "analysis of variance," was developed by R. A. Fisher (1,2).

In other words, the agronomists were less interested in the agricultural conclusions of the paper than in brandishing the new method of analysis. Their allegiance was to the method, not to the data, providing fertile ground for overlooking a mistake, especially since they did not have the benefit of the powerful visualization methods now available.

The cumulative evidence, all of it circumstantial, is nevertheless convincing. The agronomists appear to be guilty of a missed finding.

6.5 *Visualization and Probabilistic Inference*

The barley example speaks forcefully to the importance of visualization as part of the process of data analysis. Data analysis without visualization, even with methods as brilliant as Fisher's, risks missing important happenings in data, and risks missing ways of simplifying the structure of data, and risks the use of probabilistic methods whose assumptions are not supported by the data. The agronomists, using rote data analysis, missed an important happening in the data — a mistake.

The barley data also teach us that tools matter. Two data analysts of exceptional acuity — Francis J. Anscombe [3, 4] and Cuthbert Daniel [29] — analyzed the data. In the 1960s and 1970s, both contributed pioneering methods for diagnosing the performance of fits to data and for judging how well data satisfy assumptions of probabilistic inference. Both were strong advocates of visualization well before it was fashionable, and established a legacy that continues into this book. Both had deep intuition for the processes of data analysis. Both missed the significance of the Morris reversal in their analyses. Tools matter, and many of the powerful visualization tools available to us today were not available at the time of their analyses, either as software or even as ideas.

Anscombe treated the years and sites as a single categorical variable with $6 \times 2 = 12$ categories — Waseca-1931, Waseca-1932, Morris-1931, Morris-1932, and so forth — giving up on distinguishing years since it appeared to him that year and site were less interesting than variety.

It also reduced the number of factors from three to two, which made the data more conducive to visualization given the tools Anscombe had available at the time. But this merging made it nearly impossible for the Morris anomaly to appear.

Daniel analyzed the barley data using tables but no graphs. He discovered the aberrant data for Velvet at Grand Rapids and suspected an error: "These facts suggest that the two entries [the 1931 and 1932 Velvet yields at Grand Rapids] have been interchanged in error." The immediate suspicion came from a vast experience in analyzing experimental data that had taught him blunders are common. But for Morris he missed the information that triggers suspicion. He did see that the 1932 year effect exceeded the 1931 year effect. But he failed to discover that the negative of the Morris year effect fits neatly into the pattern of the year-site interaction at the other sites. So his suspicious nature was not aroused, and he wrote simply, "The yearly differences were consistent for four locations but were reversed for location 2 [Morris]." The poverty of tabular presentation did not allow even an exceptional data analyst to see the pattern in the reversal that triggers suspicion.

Visualization is a necessary part of data analysis. Tools matter.